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The Editors

Physical Review D

Dear Editors,

I am submitting the manuscript “*Static response and dynamical identifiability in free-fall tests: finite-order boundaries and a pulsar-triple application*” for consideration as a regular article in *Physical Review D*, in the area of gravitation and tests of gravity.

Free-fall tests with the pulsar triple PSR J0337+1715 constrain body-dependent couplings with great precision, and they are usually read through static sensitivities of the bodies. The manuscript asks a sharper question: when a body has an internal state that relaxes on orbital timescales, what can free-fall timing data identify beyond a static description, and under which conditions?

The main results are:

- A finite-order static reference space, in which an electric-tidal quotient has five representatives through weight four.
- An interpolation boundary: a nonzero single pole can be matched at K positive frequencies by real derivative coefficients of degree N if and only if $N \geq 2K - 1$. For a positive reciprocal relaxation spectrum, two calibrated complex samples instead identify one observable relaxation time through a moment-variance equality, while finite precision cannot separate arbitrarily close poles.
- A damped scalar-charge realization with reciprocal forces and a controlled one-pole reduction, applied conditionally to a stored J0337 timing analysis. Its nuisance-space dependence and the failed empirical-promotion gates are reported rather than hidden.
- A worked calculation for the inner white dwarf of the triple. A scalar pulse leaves a transient retarded monopole signal in the star’s displaced matter. At orbital timescales its structural response is small, and a layer-by-layer thermal-relaxation estimate puts the lagged part at 4×10^{-9} to 3.3×10^{-7} of that term. The corresponding timing sensitivity is not computed, so the paper states a scale comparison, not an exclusion.

I believe the paper suits *Physical Review D* because it connects the effective-field-theory description of extended bodies with the inference practice of pulsar tests of gravity. It states precisely which conclusions are proven, which rest on stored computations, and which remain conjectural: every substantive claim carries one of four explicit labels (Proven, Imported from prior work, Counterexample candidate, Conjectural). The paper does not claim a new empirical bound on violations of the strong equivalence principle.

The code, numerical records and SHA-256-bound manifests are available in the public repository <https://github.com/lpaiu-cs/self-mass-unobservability>; the Data and Code Availability section states what is and is not deposited. AI agents were used substantively in the research and in drafting the text. This use is disclosed in the manuscript (Section 7), following the APS policy on AI tools.

The manuscript has not been published and is not under consideration elsewhere. I have no competing interests to declare.

Sincerely,

Juneyoung Kim